The SEFI Framework: Foundations of the Single–Electron Mode

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Abstract

This paper presents the core idea behind the SEFI framework: the electron is not a point object or a probabilistic cloud, but a geometric mode with internal structure. Everything in SEFI grows out of this single assumption. Earlier attempts to write this down took longer because I relied on AI tools and the results never sounded like me. This version is written directly and intentionally, in my own voice, to establish a clear foundation for the rest of the work.

1 Introduction

SEFI begins with one idea: the electron is a geometric mode defined on a manifold with curvature, resonance, and coherence structure. If this mode is accepted, the rest of the framework follows naturally. If it is not accepted, nothing else in SEFI can be understood.

The purpose of this paper is to explain why the single–electron mode works and to present the minimal mathematical structure needed to describe it.

2 Why the Single–Electron Mode Works

The starting point is simple. If a single electron could move through all available directions in spacetime, both forward and backward in time, and occupy every geometric position along its worldline, then it would not appear as a particle. It would appear as a field.

This is a geometric consequence, not a philosophical claim.

A particle moving through time traces a worldline. A particle moving through all directions traces a dense set of worldlines. A particle moving through all possible spacetime paths

generates a world–tube. When the density of these paths becomes high enough, the object no longer presents itself as a localized excitation. It presents itself as a continuous field.

In SEFI, this field is the *reality field*.

The electron is not spread out because of probability. It is spread out because it occupies all available geometric directions. To a three–dimensional observer, this extended mode is indistinguishable from a field.

3 Mathematical Formalism

3.1 Electron Mode

Define the electron mode as a geometric field

$$\psi : \mathcal{M} \to \mathbb{C}, \tag{1}$$

with internal geometric structure.

3.2 Curvature Operator

The curvature operator describes geometric tension within the mode:

$$\mathcal{C}[\psi] = g^{\mu\nu}\nabla_\mu\nabla_\nu\psi. \tag{2}$$

3.3 Resonance Operator

The resonance operator describes the intrinsic oscillatory behavior:

$$\mathcal{R}[\psi] = \omega^2\psi. \tag{3}$$

3.4 Coherence Field

The coherence field measures stability:

$$\Phi = \psi^*\psi. \tag{4}$$

3.5 Fundamental SEFI Equation

The core SEFI equation is

$$\mathcal{C}[\psi] + \mathcal{R}[\psi] = \Phi. \tag{5}$$

This equation defines the balance between curvature, resonance, and coherence that stabilizes the electron mode.

4 Discussion

This paper establishes the minimal mathematical foundation of SEFI. All later constructions—warp–metric behavior, multi–mode interactions, biological manifolds, and system–level geometry—depend on accepting the electron as a geometric mode with internal structure.

Earlier attempts to formalize this were slowed by the use of AI tools, which produced inconsistent phrasing and unstable notation. This version is written directly and intentionally, with the goal of providing a stable reference for future work.

5 Conclusion

The single–electron mode is the foundation of SEFI. By defining the curvature operator, resonance operator, coherence field, and the fundamental SEFI equation, this paper provides the core mathematical structure needed to understand the rest of the framework.